

The Weak-EKR Difference-Set Condition for Irreducible Subgroups of $\mathrm{GL}(3, 2)$

Xiaojian Zeng

PBC School of Finance, Tsinghua University, Beijing 100190, China

xiaojianzeng@foxmail.com

Abstract

Let $V = \mathbb{F}_2^3$. We classify, up to conjugacy in $\mathrm{GL}(3, 2)$, the irreducible subgroups L and decide whether

$$A - A \subseteq \bigcup_{\ell \in L} \ell(W) \implies |A| \leq |W|$$

holds for every subspace $W \leq V$ and every subset $A \subseteq V$. There are exactly three irreducible conjugacy classes, of orders 7, 21, and 168, and none satisfies the implication. The same explicit pair $W = \langle e_1 \rangle$ and $A = \{0, e_1, e_2\}$ is a counterexample for all three displayed representatives. Completeness is certified by an exact enumeration of all 179 subgroups, a partition into 15 ambient conjugacy classes, and direct invariant-subspace tests. A separate implementation reconstructs and verifies every finite calculation.

1 Statement and context

For a subgroup $L \leq G := \mathrm{GL}(V)$ and a subspace $W \leq V$, put

$$\mathcal{U}_L(W) = \bigcup_{\ell \in L} \ell(W).$$

We call L irreducible when its only invariant subspaces are 0 and V . The property under consideration is

$$\forall W \leq V \ \forall A \subseteq V, \quad A - A \subseteq \mathcal{U}_L(W) \implies |A| \leq |W|. \quad (1)$$

Changing from a left-action convention to the inverse/right-action convention does not change $\mathcal{U}_L(W)$, because inversion permutes L .

Frelih, Hujdurović, and Kutnar derive this difference-set condition from the weak Erdős–Ko–Rado property, make rank one automatic, characterize rank two, and ask for an analogous characterization in ranks at least three [2, Proposition 2.6, Corollary 2.7, Problem 2.9]. The following theorem resolves the smallest higher-rank finite slice. It does not assert a characterization for general rank or field size.

Theorem 1. *Up to conjugacy in $G = \mathrm{GL}(3, 2)$, the irreducible subgroups have orders 7, 21, and 168. Every one of these three conjugacy classes fails (1). In particular, no irreducible subgroup of $\mathrm{GL}(3, 2)$ satisfies (1).*

The surrounding classification theory for transitive finite linear groups is substantial; see, for example, Ernst and Schmidt [1]. Nothing from that classification is needed below: at order 168 it

is shorter to retain a complete exact census. Exact symmetry-reduced enumeration with independently checked predicates has also been used in other finite extremal problems [4]; explicit-generator arguments provide a complementary certification style in finite permutation groups [3]. These last two citations are methodological comparisons only and do not support any assertion about (1).

2 Complete subgroup and irreducibility census

Vectors are encoded by the integers $0, \dots, 7$, with bit i the coefficient of e_{i+1} . A matrix is encoded by the ordered triple of images of the basis vectors. Thus matrix application and multiplication use only bitwise exclusive or. Exhausting the linearly independent ordered column triples gives

$$|G| = (2^3 - 1)(2^3 - 2)(2^3 - 2^2) = 168.$$

Starting with the trivial subgroup, repeatedly adjoin each element of G to each subgroup already found. At termination this produces a family $\mathcal{F}$ of 179 distinct subsets. Every member is checked to contain the identity and to be closed under products and inverses. The following elementary criterion makes the census proof-carrying rather than heuristic.

Lemma 1 (single-adjunction saturation). *Let G be finite and let $\mathcal{F}$ be a family of subgroups containing $\{1\}$. If $\langle H, g \rangle \in \mathcal{F}$ for every $H \in \mathcal{F}$ and $g \in G$, then $\mathcal{F}$ contains every subgroup of G .*

Proof. If $K \leq G$, list its elements as $g_1, \dots, g_r$. Beginning with $H_0 = \{1\}$, saturation gives $H_i = \langle H_{i-1}, g_i \rangle \in \mathcal{F}$ inductively. Since $H_r = K$, the subgroup K is listed. $\square$

For the computed family, all $179 \cdot 168 = 30,072$ instances of the saturation hypothesis were checked. The subgroup counts are

order	1	2	3	4	6	7	8	12	21	24	168
number	1	21	28	35	28	8	21	14	8	14	1

Exact conjugation by all elements of G partitions this family as follows. The orbit sizes sum to 179, so the rows form a disjoint exhaustive partition.

order	1	2	3	4	4	4	6	7	8	12	12	21	24	24	168
orbit size	1	21	28	7	7	21	28	8	21	7	7	8	7	7	1
irreducible?	no	no	no	no	no	no	no	yes	no	no	no	yes	no	no	yes

There are seven lines and seven planes in V . For each of the 15 representatives, all 14 proper nonzero subspaces were acted on by every subgroup element. Exactly the representatives of orders 7, 21, and 168 have no invariant line or plane. Since an invariant proper nonzero subspace in dimension three has one of these two dimensions, this test is equivalent to irreducibility. Irreducibility is conjugacy invariant, so the census proves the first assertion of Theorem 1.

3 Representatives and a common counterexample

Here are representatives of the three irreducible classes. The matrices on each line generate the indicated subgroup:

$$\begin{aligned} L_7 &= \left\langle \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \right\rangle, \\ L_{21} &= \left\langle \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \right\rangle, \\ L_{168} &= \left\langle \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \right\rangle = G. \end{aligned}$$

Closing these displayed generators under multiplication gives respectively 7, 21, and 168 elements. Direct application of those elements gives, in every case,

$$L_i e_1 = V \setminus \{0\}, \quad i \in \{7, 21, 168\}. \quad (2)$$

Completion of the proof of Theorem 1. For any one of the displayed groups let

$$W = \langle e_1 \rangle = \{0, e_1\}, \quad A = \{0, e_1, e_2\}.$$

Equation (2) gives $\mathcal{U}_{L_i}(W) = V$. Since the characteristic is two,

$$A - A = A + A = \{0, e_1, e_2, e_1 + e_2\} \subseteq V, \quad |A| = 3 > 2 = |W|.$$

Thus (L_i, W, A) violates (1) for each of the three exhaustive irreducible classes. $\square$

4 Exact certificate and reproducibility

The primary certificate serializes all 168 matrices, all 179 subgroups, their generators, the 15 conjugacy orbits, every invariant-subspace result, and the three witnesses. As a redundant test, all 16 subspaces W and all 256 subsets A were checked for each irreducible representative, for $3 \cdot 16 \cdot 256 = 12,288$ exact pairs. Each class has 2,184 violations; only one per class is needed for the theorem.

A standalone verifier imports neither generating program. It reconstructs the 168 matrices and multiplication table from the definition, checks 45,473 within-subgroup products, repeats the 30,072 saturation tests, reconstructs the conjugacy partition (checking 50,400 conjugated elements), retests every line and plane, repeats all 12,288 difference-set tests, and checks the displayed witnesses. All arithmetic is exact over $\mathbb{F}_2$; there is no random sampling or floating-point step. The scripts and JSON certificates are supplied with the article. In particular, the supplement contains `enumerate_g132_subgroups.py`, `test_weak_ekr.py`, and the independently written `verify_certificates.py`, together with their subgroup, classification, and verification JSON outputs.

The comparison with prior work was deliberately focused. The inspected weak- EKR source does not state this finite classification, but the search does not justify a priority claim. The contribution asserted here is only the complete, self-contained resolution of the stated $(d, p) = (3, 2)$ problem.

Computational and AI assistance. The finite search, certificate generation, independent replay, and preparation of this manuscript used computational and AI assistance. AI-generated text or confidence assessments are not treated as mathematical evidence. The result rests on Lemma 1, the displayed exact witness, and the deterministic finite certificates described above; a separately implemented verifier rechecks all decisive relations.

References

- [1] Alena Ernst and Kai-Uwe Schmidt. Transitivity in finite general linear groups. *Mathematische Zeitschrift*, 307(3):52, 2024.
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